

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (CL) Examination

December 2013 - January 2014

CL 701 Finite Element Analysis

Date: 01.01.2014, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator/ computer terminal with MS Excel spreadsheets developed is allowed.

SECTION - I

- Q - 1 For a one-dimensional finite element, with two nodes and one displacement degree of freedom at each of the nodes, derive in a step-by-step manner shape function matrix $[N]$ by assuming suitable polynomial displacement function. [07]
- Q - 2 Using any one of (i) Galerkin's weighted residual method, or (ii) potential energy approach, for the bar hanging under its own weight as shown in Figure 1, determine the nodal displacements at selective points along the length of the bar. Let bar cross-sectional area, $A = 1 \text{ in.}^2$, modulus of elasticity, $E = 30 \times 10^6 \text{ psi}$, and weight density, $\rho_w = 0.283 \text{ lb/in.}^3$. Here, lb, in., and psi respectively denote pounds, inch, and pounds per square inch. [14]

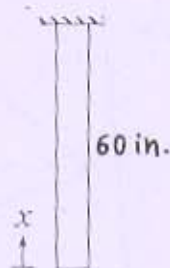


Figure 1
OR

- Q - 2 For the tapered bar shown in Figure 2, the cross-sectional areas are $A_1 = 1 \text{ in.}^2$ and $A_2 = 0.5 \text{ in.}^2$ at each end. Using higher order isoparametric formulation, obtain the displacements in the bar at 5 in. interval along the length of the bar. Here, lb, in., and psi respectively denote pounds, inch, and pounds per square inch. [14]

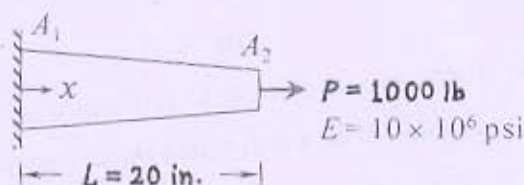


Figure 2

- Q - 3 For the bar hanging under its own weight as shown in Figure 1, determine the strains, [14]
and stresses at selective points along the length of the bar. Remaining data is to be
considered exactly the same from the concerned question.

OR

- Q - 3 For the tapered bar shown in Figure 2, obtain the strains and stresses in the bar at 5 in. [14]
interval along the length of the bar. Remaining data is to be considered exactly the
same from the concerned question.

SECTION - II

- Q - 4 For a linear elastic spring of stiffness 90 N/mm, which is in equilibrium under a [07]
tensile force of 5000 N, determine the minimum potential energy and corresponding
deformation in the spring.
- Q - 5 For a three-dimensional (3-D) tetrahedron element shown in Figure 3, the coordinates [14]
of the nodes are given as 1(0,0,2); 2(2,0,0); 3(0,2,0); 4(0,0,0). Assume that the nodal
displacements have been determined to be:

$$\begin{array}{llll} u_1 = 0.005'' & u_2 = 0.001'' & u_3 = 0.005'' & u_4 = -0.001'' \\ v_1 = 0.000'' & v_2 = 0.000'' & v_3 = 0.000'' & v_4 = 0.000'' \\ w_1 = 0.000'' & w_2 = 0.001'' & w_3 = 0.000'' & w_4 = 0.005'' \end{array}$$

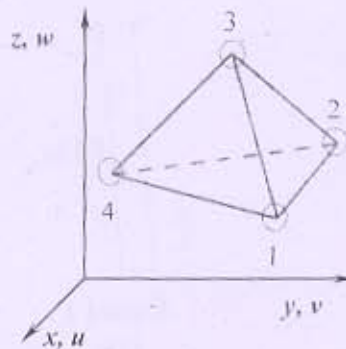


Figure 3 (Axes/Coordinates Indicative, Data Given in Question Statement)

Determine the strains in the element. Let modulus of elasticity, $E = 30 \times 10^6$ psi and
Poisson's ratio, $\nu = 0.3$. The " symbol denotes inch unit, and psi denote pounds per
square inch, respectively.

OR

- Q - 5 Determine the nodal displacements for the 1" uniformly thick plate of dimensions [14]
(20"×10") as shown in Figure 4 with a distributed shear load acting all along the right
edge uniformly, $s = 1000$ lb/in. Let modulus of elasticity, $E = 30 \times 10^6$ psi and

Poisson's ratio, $\nu = 0.3$. The " symbol denoted inch unit, lb and psi denote pounds and pounds per square inch, respectively.

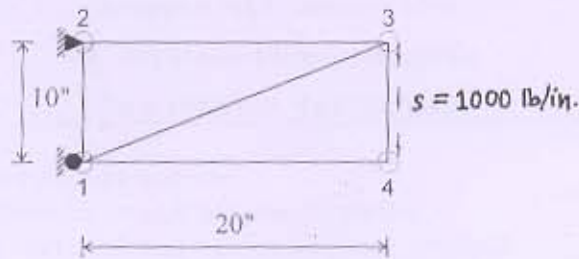


Figure 4

- Q - 6 For a three-dimensional (3-D) tetrahedron element shown in Figure 3, determine the stresses in the element. Remaining data is to be considered exactly the same from the concerned question. [14]

OR

- Q - 6 Determine the element stresses, including principal stresses, for the plate shown in Figure 4. Remaining data is to be considered exactly the same from the concerned question. [14]
